

SSGMCE/PRM/52-B		SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGG.		LABORATORY MANUAL	
SSGMCE	PRACTICAL EXPERIMENT INSTRUCTION SHEET				
	EXPERIMENT TITLE : Simulation of a Simple Intelligent Agent Using Python				
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01 AIM: Write a Program to simulate a simple intelligent reflex agent operating in a vacuum cleaner environment of two rooms (A and B)

02 SCOPE: To understand and implement basic Artificial Intelligence concepts by simulating simple intelligent agents and analyzing their behavior in different task environments. Gain hands-on experience in implementing AI concepts using Python.

03 FACILITIES: Computer System with Python

04 THEORY:

An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment using actuators. The agent's goal is to maximize its performance measure based on its percepts.

In this experiment, a Simple Reflex Agent is simulated in a Vacuum Cleaner Environment consisting of two rooms (A and B). The agent decides actions based only on the current percept without considering past percepts.

Task Environment Characteristics

- Fully observable
- Deterministic
- Sequential
- Static
- Discrete

Algorithm

- 1) Initialize the environment with two rooms (A and B) having random cleanliness status.
- 2) Place the agent randomly in one of the rooms.
- 3) Sense the current room's status.
- 4) If the room is dirty, perform the clean action.
- 5) If the room is clean, move to the adjacent room.
- 6) Repeat the above steps for a fixed number of iterations.
- 7) Display the agent's actions and updated environment status.

05 PROGRAM:

```
import random
```

```
# Initialize environment
```

```
rooms = {  
    'A': random.choice(['Dirty', 'Clean']),  
    'B': random.choice(['Dirty', 'Clean'])  
}
```

```
agent_location = random.choice(['A', 'B'])
```

```
def simple_reflex_agent(location, status):
```

```
    if status == 'Dirty':  
        return 'Clean'  
    elif location == 'A':  
        return 'Move Right'  
    else:  
        return 'Move Left'
```

```
print("Initial Environment Status:", rooms)  
print("Initial Agent Location:", agent_location)  
print("-" * 40)
```

```
for step in range(5):
```

```
    current_status = rooms[agent_location]  
    action = simple_reflex_agent(agent_location, current_status)
```

```
    print(f"Step {step + 1}: Agent at Room {agent_location}, Status = {current_status}")  
    print("Action:", action)
```

```
    if action == 'Clean':  
        rooms[agent_location] = 'Clean'  
    elif action == 'Move Right':  
        agent_location = 'B'  
    elif action == 'Move Left':  
        agent_location = 'A'
```

```
    print("Updated Environment:", rooms)  
    print("-" * 40)
```

06 OUTPUT:**Sample output:**

Initial Environment Status: {'A': 'Dirty', 'B': 'Clean'}

Initial Agent Location: B

Step 1: Agent at Room B, Status = Clean

Action: Move Left

Updated Environment: {'A': 'Dirty', 'B': 'Clean'}

Step 2: Agent at Room A, Status = Dirty

Action: Clean

Updated Environment: {'A': 'Clean', 'B': 'Clean'}

Step 3: Agent at Room A, Status = Clean

Action: Move Right

Updated Environment: {'A': 'Clean', 'B': 'Clean'}

Step 4: Agent at Room B, Status = Clean

Action: Move Left

Updated Environment: {'A': 'Clean', 'B': 'Clean'}

Step 5: Agent at Room A, Status = Clean

Action: Move Right

Updated Environment: {'A': 'Clean', 'B': 'Clean'}

Note: Output may vary due to random initialization of environment and agent location.

07 CONCLUSION:

the simulation of a simple reflex agent was successfully implemented using Python. This experiment demonstrates the fundamental concepts of intelligent agents, agent–environment interaction, and task environment classification.